

2. Difference between a method declaration and a method body

The method declaration is the first line (name, return type, parameters).

The method body is the code inside the braces that actually runs.

3. Keyword used to change a method's access level

Access modifiers like public, private, or protected.

4. Another word for access level

Visibility.

5. Scope of each variable

- **var1:** Only inside main.
- **var2:** Only inside the loop in main.
- **var3:** Only inside method1.
- **var4:** Only inside the loop in method1.

6. Method declarations

```
public static int getVowels(String word)
public static int extractDigit(int number)
public static String insertString(String text, int value)
```

7a. How the compiler tells methods apart

By the method name and its parameter list (the method signature).

7b. Can two methods have the same name?

Yes. That's method overloading, as long as their parameters are different.

8a. What a return statement does

It sends a value back to the caller.

8b. How many values can it return?

Just one.

8c. Difference between returning and non-returning methods

Returning methods have a return type and must return a value.

Void methods use void and return nothing.

9. Error in the code

The method is declared void but tries to return(5);.

Void methods can't return a value.

11. True/False

- a) True – breaking tasks into smaller methods is abstraction.
- b) False – a method call does not declare a method.
- c) False – void methods return nothing.
- d) False – access modifiers set visibility, not return type.
- e) False – static means the method belongs to the class.
- f) False – parameters go in parentheses.
- g) False – local variables only work inside their method.
- h) True – reference types can be changed by a method.
- i) False – overloading just means same name, different parameters.
- j) True – return sends back a value.
- k) True – a precondition is what must be true before the method runs.
- l) True – a postcondition is what's guaranteed after the method runs.